



Supervised ML Workshop

DIT Pimpri



A workshop by Atharv, Ayush, Yash

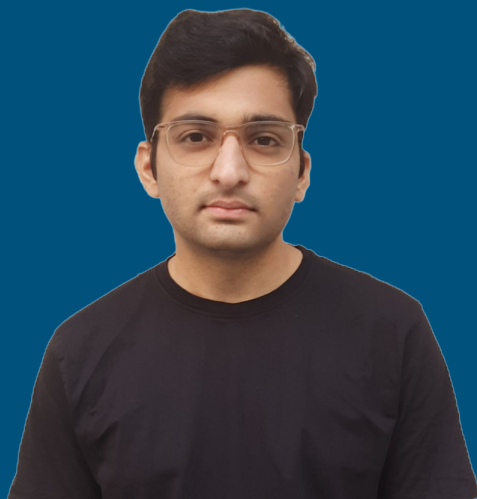


Agenda for Workshop

Supervised Machine Learning

1. Overview of ML
2. Linear Regression
3. Logistic Regression
4. Decision Trees
5. Ensembles ←





Atharv

AI/ML Developer

GDG DIT - Machine Learning Lead

Works with the AI team at
ElevateTrust.AI

Always learning, building, and
sharing

Passionate about AI, ML, Generative
AI, EdgeAI



Ayush

AI/ML Developer

GDG DIT - Machine Learning Co-Lead

Turning data into insights with ML & AI, one model at a time.

Building apps where Machine Learning meets real-world impact.

Speedcuber — learned algorithms from cubes, patience (and suffering) from code.



Yash

AI/ML Developer

GDG-DIT Research Lead

Ex CP Member Lead

Ex Intern Accenture

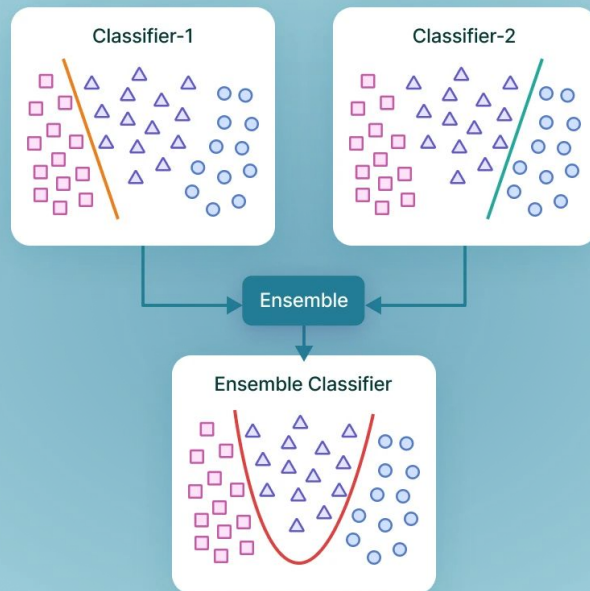
Passionate about innovative tech

Forever Student

ENSEMBLE LEARNING

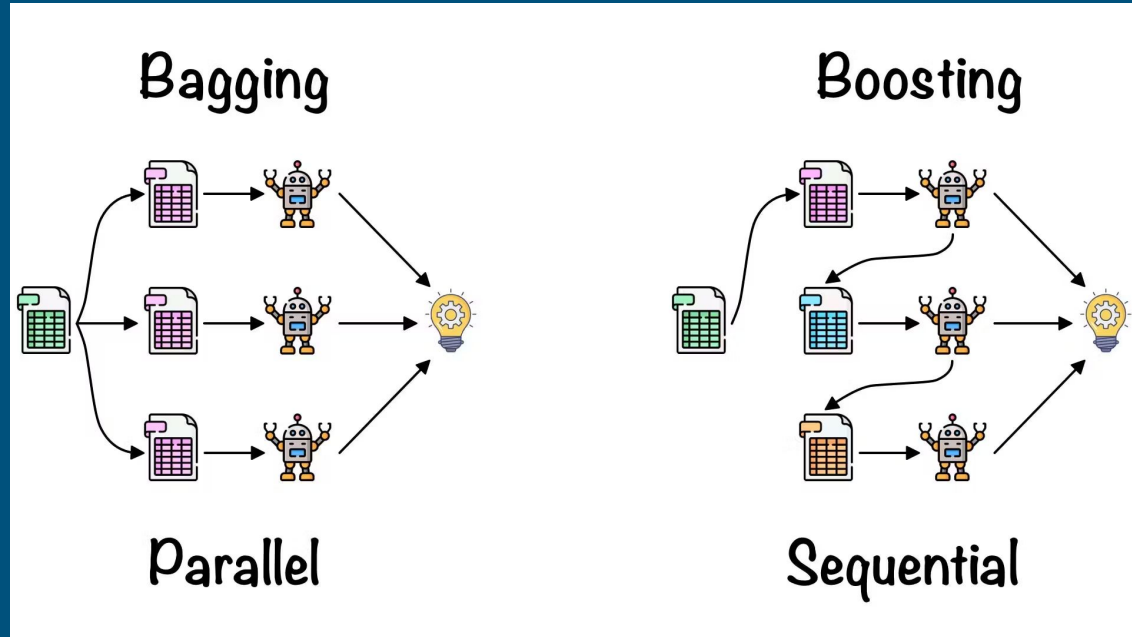
Ensemble learning ek technique hai jisme multiple models (jinhe base learners kaha jaata hai) ko combine karke ek final prediction banaya jaata hai, jo ek single model se zyada accurate hota hai.

Ye bilkul waise hi hai jaise kai experts se opinion lena aur unke combined decision par rely karna — jisse result zyada accurate aur stable milta hai.



Types of Ensemble Learning

1. Bagging (Bootstrap Aggregating)
2. Boosting

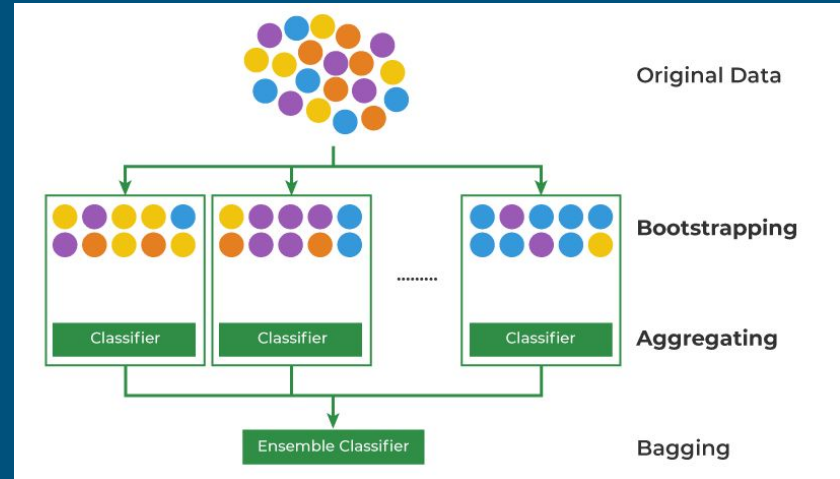


BAGGING

Bagging ka main idea hai multiple models ko alag-alag random data samples par train karna.

Har model thoda-thoda different data dekhta hai, isliye sabke predictions alag hote hain.

Phir un sabke results ko average (regression) ya majority vote (classification) karke final output nikalte hain.

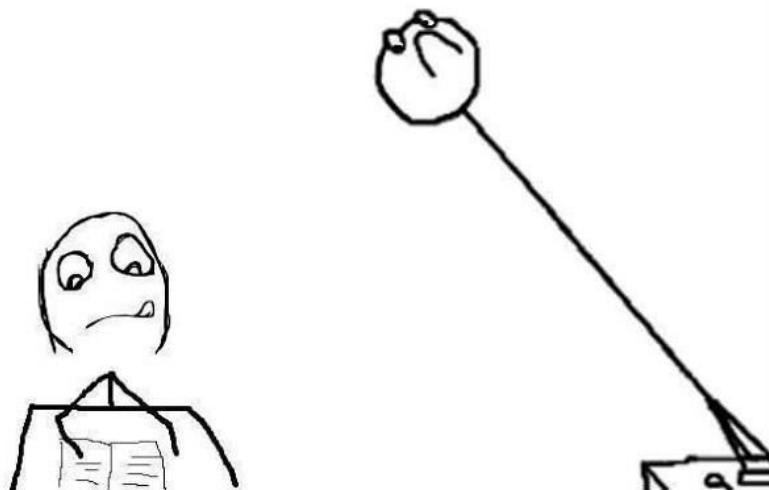




Teachers Call It Cheating

We call it teamwork

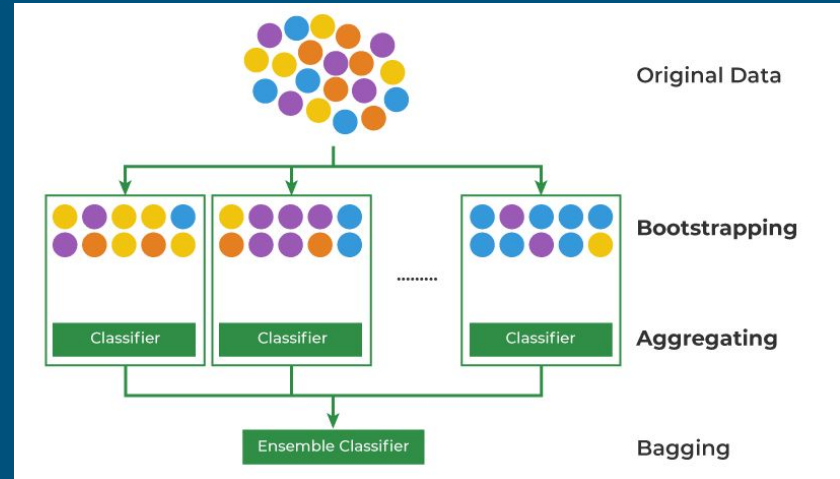
During a test



BOOSTING

Boosting ka concept thoda different hai.
Yahan models ek ke baad ek (sequentially)
train hote hain.

Har naya model pehle wale model ke
galtiyon (errors) par focus karta hai.

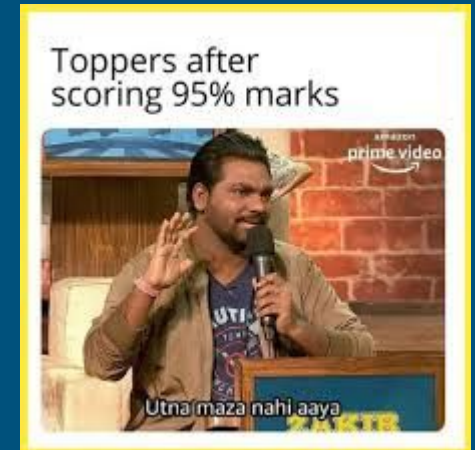
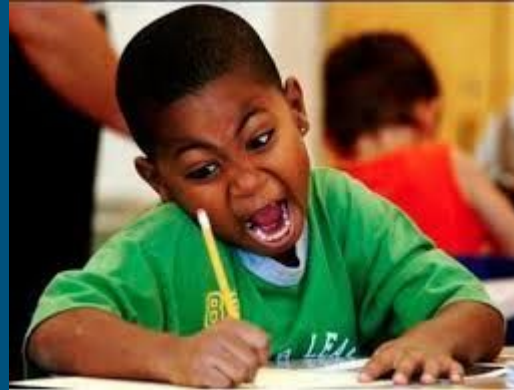
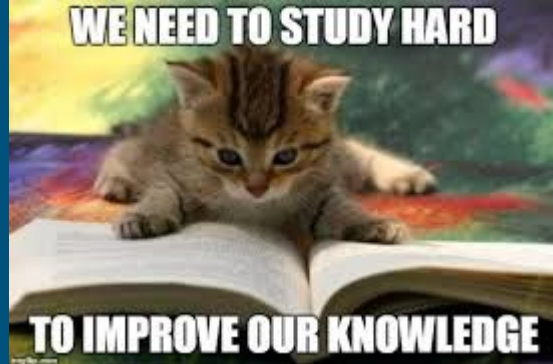


**WHEN YOU REALIZE YOUR
JEE RANK IS YOUR ROLL
NUMBER**



	$\frac{a + b}{2}$
	$\frac{1}{n} \sum_{i=1}^n x_i$
	$\frac{1}{b-a} \int_a^b f(x) dx$

	Dispersion Force
	Dipole-Dipole Interaction
	Hydrogen Bonding



Fail → Learn → Fix → Repeat → Topper Ban Ja!

Random Forest

Random forests bag deep decision trees with a slight twist. While growing each tree, at each split (question-answer pair), instead of considering all d features, it randomly samples a subset of m features to choose from. m is typically around $\sqrt{d}$. In this way, the resulting trees are uncorrelated. Since the trees are grown independently on their own bags, the algorithm can be run in parallel.



XGBOOST

XGBoost builds a model by creating a team of decision trees. The first tree makes an initial prediction. Then, every subsequent tree is trained to fix the mistakes of the one before it. XGBoost is 'Extreme' because it does this in a very fast, clever, and robust way, which is why it's one of the most popular and effective models in machine learning today

1. Start with a single base prediction (e.g., the average of the target variable).
2. For each iteration t :
 - a. Calculate the gradients (g_i) and Hessians (h_i) for each data point based on the current prediction.
 - b. Build a new tree (f_t) by greedily finding the best splits that maximize the "Gain" score.
 - c. Calculate the optimal leaf scores (w_j) for this new tree's terminal nodes.
 - d. Update the model's overall prediction: $\hat{y}_i(t) = \hat{y}_i(t-1) + \eta \cdot f_t(x_i)$, where η is a learning rate to shrink the contribution of each new tree.

Repeat step 2 for the desired number of trees. The final prediction is the sum of the initial prediction and all the trees added.

Other famous models

- AdaBoost
- Gradient Boosting Machine (GBM)
- Light GBM
- CatBoost

Assignment

1. Ask doubts learn more



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Thank You